

Math Kangaroo 26 Summer

Grade 1-2

Lecturer: Zong

What is Math Kangaroo?

- The Math Kangaroo competition is a test



Different shapes
Different flavors
Surprising by what is in side



pencil is your FRIEND

Math Kangaroo Style

- Mia has a small, square box that holds exactly 4 chocolates in a 2x2 grid. She has 2 white chocolates and 2 dark chocolates.
- She wants to put them in the box so that no two chocolates of the same color are sitting next to each other in the same row or column.
- How many different ways can she arrange the 4 chocolates?

4颗巧克力的排列挑战



答案：
2种本质不同的排法

2白 | 2黑 | 不能相邻同色

3

4 step method

绝密档案：数学小侦探的 4 个大招！

Top Secret: 4 Steps for Math Detectives!

1 找线索 (Find Clues)

拿起你的放大镜，找出数字和问题！



圈一圈，划一划！
Read it, Circle it!

2 画图纸 (Draw a Picture)

字太多看不懂？用魔法把它变成一幅画！
画圈圈、画火柴人。



画一画，变魔法！
Draw a sketch!

3 破密码 (Crack the Code)

看着你的画，写出算式，找出藏着的答案！



列算式，破密码！
Write the numbers, crack the code!

4 回头看 (Look Back)

别急着跑！把答案放回故事里。



回头看，不遗漏！
Look back, avoid the trap!

★ 侦探宣誓 (Detective Pledge): 我承诺：做题不靠猜，画图最厉害！
I promise: No guessing without drawing!

侦探签名
Detective Signature

18. Ann buys 3 chocolate bars. Tom buys 5 chocolate bars. Tom paid 8 euro more than Ann. What is the cost of 1 chocolate bar?

小安买了 3 条巧克力棒，小汤买了 5 条巧克力棒。小汤比小安多付了 8 欧元。
请问一条巧克力棒的价格是多少？

(A) 1 euro 1 欧元

(B) 2 euro 2 欧元

(C) 3 euro 3 欧元

(D) 4 euro 5 欧元

(E) 5 euro 5 欧元

14. There are 12 students in a line. Counting from the left, Warren is fifth in the line. Counting from the right, Victor is fourth in the line. How many students are between Warren and Victor?

12 个学生站成一排。从左边开始数，小沃是第五个。从右边开始数，小维是第四个。问小沃和小维中间有多少个学生？



(A) 1

(B) 2

(C) 3

(D) 4

(E) 5

2025

23. Each time a coin is put in the machine, a ball falls at random from the bottom row. What is the smallest number of coins Bahar must have to be sure that she will get a white ball?

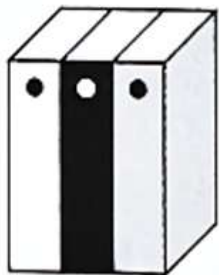
每次往机器里投一枚硬币，就会随机有一个球从底部掉落。问小巴至少要投多少枚硬币才能确保能拿到一个白球？

- (A) 6 (B) 10 (C) 11 (D) 12 (E) 15



15. There were three books on Mary's shelf, as shown in the picture.

小玛的书架上有一本书，最初的位置如图所示。



● First, Mary swapped the white book and the grey book.

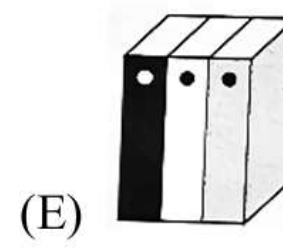
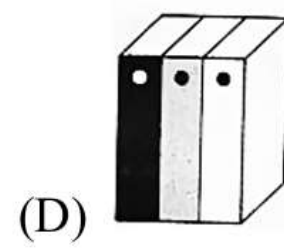
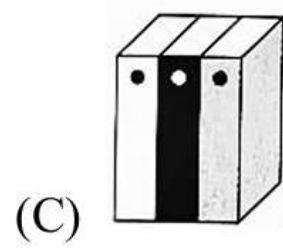
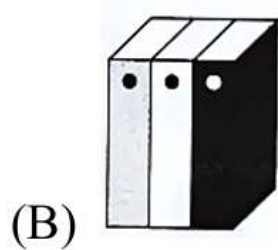
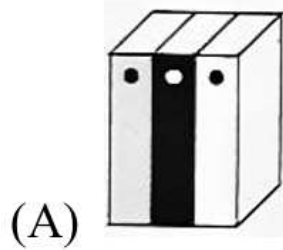
首先，小玛交换了白色书和灰色书。

● Then Mary swapped the grey book and the black book.

然后，小玛交换了灰色书和黑色书。

How are her books arranged now?

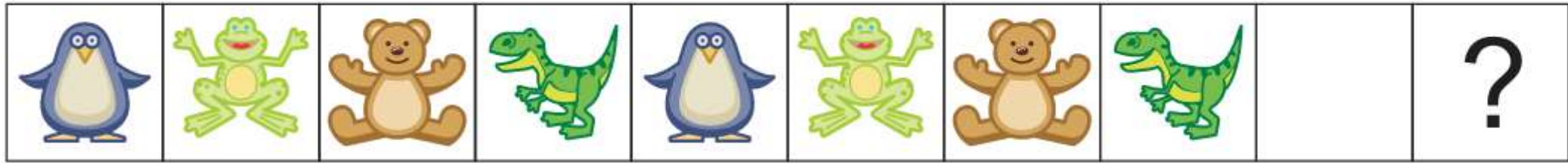
问现在她的书是和摆放的？



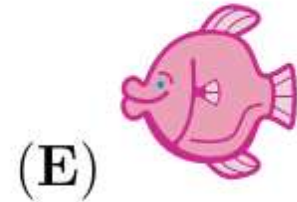
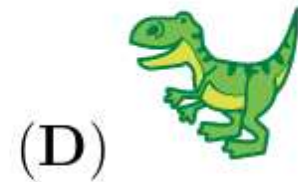
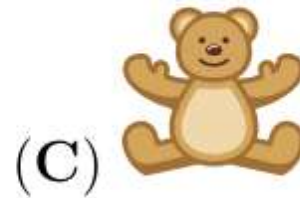
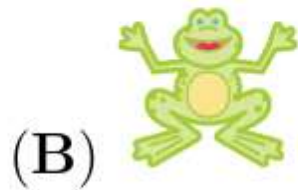
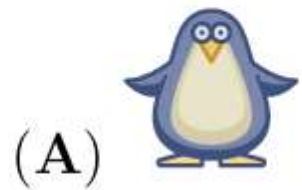
第1课 发现规律 (Patterns)

《2012》Pre-Ecolier 第5题

5. Luke repeats the same four stickers on a strip.



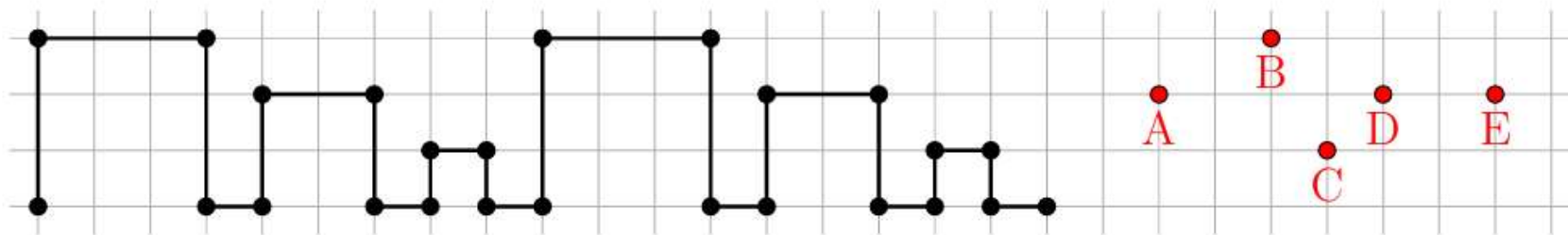
Which is the tenth sticker put by Luke?



《2018》 Level 1-2 第10题

10. Peter drew a pattern twice, as in the figure. Which point will he pass when he draws the same pattern for the third time?

如图所示，Peter 重复地画了两次同样的图案。请问，如果他把同样的图案重复第三次，会经过哪一个点？



(A) A

(B) B

(C) C

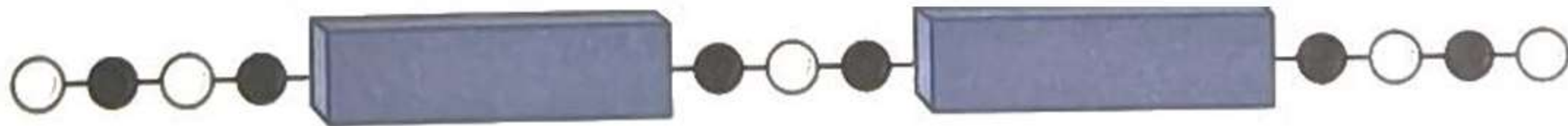
(D) D

(E) E

《2024》A级别 第15题 Bonus

17. The beads on a string alternate between black and white.

穿在绳子上的珠子黑白交替排列，如图所示。



Each block hides five beads. How many white beads are hidden in total?

每个木块遮住了 5 颗珠子。问总共有多少颗白色珠子被遮住了？

(A) 5

(B) 6

(C) 7

(D) 9

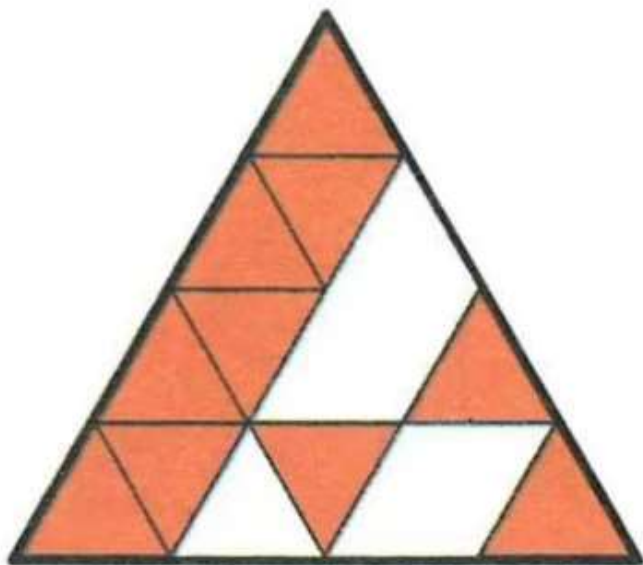
(E) 10

第2课 画图策略 (Draw it out)

《2024》A级别 第6题

- 6、 Bruno is making this large triangle using identical small triangular tiles. How many more tiles does Bruno need to complete the large triangle?

*Bruno*在用相同的小三角形瓷砖拼一个大三角形. *Bruno*还需要多少块瓷砖才能完成这个大三角形?



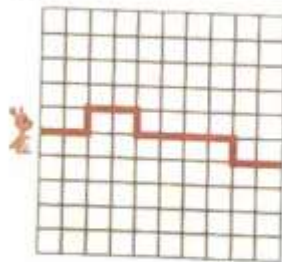
- A. 3 B. 4 C. 5 D. 6 E. 7

《2025》A级别 第5题

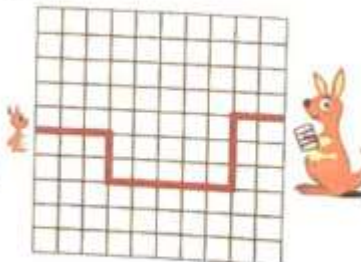
5. five baby kangaroos are hopping to their mothers along the paths shown below.
Which path is the shortest?

5 只袋鼠宝宝沿着下方所示的路径跳向它们的妈妈。问哪条路径最短？

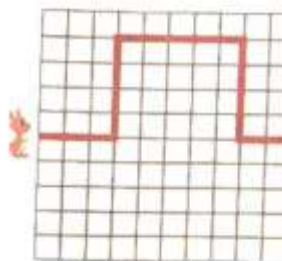
(A)



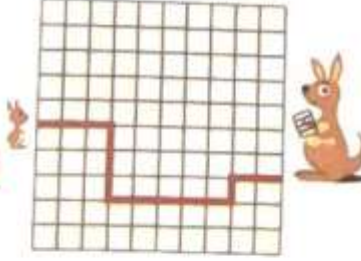
(B)



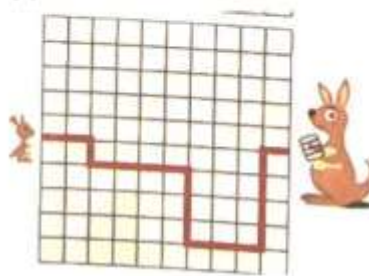
(C)



(D)

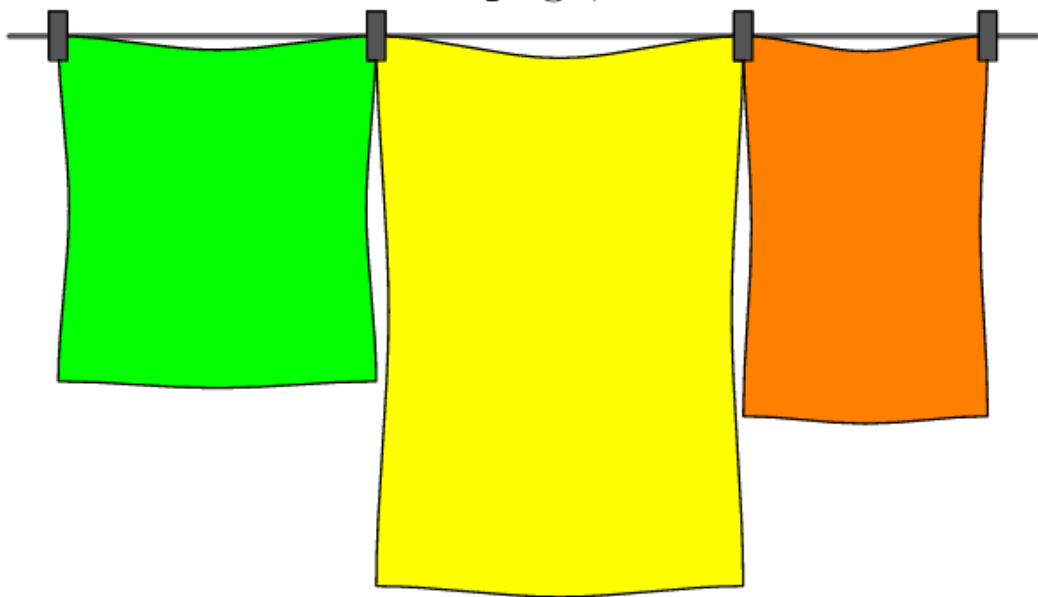


(E)



《2012》Pre-Ecolier 第10题

10. Father hangs the laundry outside on a clothesline. He wants to use as few pegs as possible. For 3 towels he needs 4 pegs, as shown. How many pegs does he need for 9 towels?



(A) 9

(B) 10

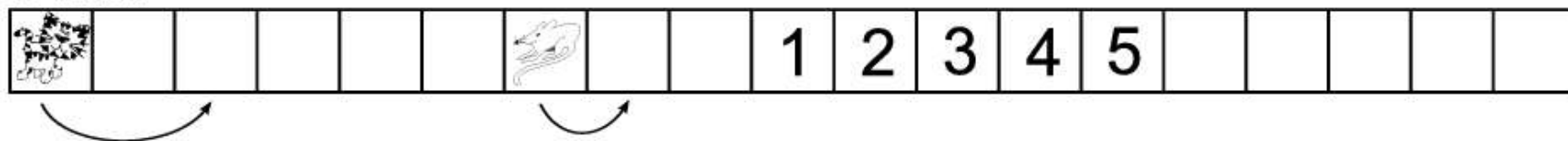
(C) 12

(D) 16

(E) 18

《2013》 Pre-Ecolier 第13题

13. Cat and Mouse are moving to the right. When Mouse jumps 1 tile, Cat jumps 2 tiles at the same time.



On which tile does Cat catch Mouse?

(A) 1

(B) 2

(C) 3

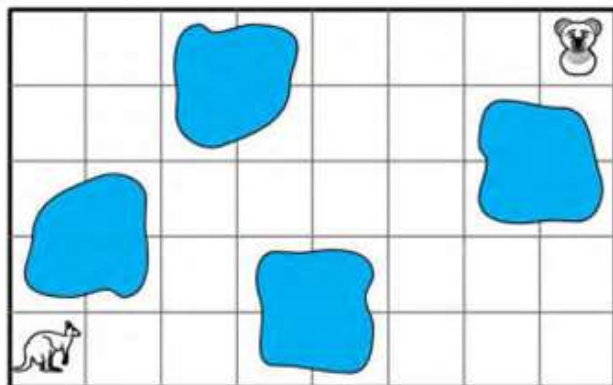
(D) 4

(E) 5

《2022》 Level 1-2 第11题

11. Kanga wants to reach the koala without going through any of the coloured squares.

dài shǔ xiǎng zhǎo kǎo lā wán dàn bù xiǎng jīng guò rèn hé yī gè cǎi sè fāng kuài
袋鼠想找考拉玩，但不想经过任何一个彩色方块。



Which route could she take?

tā kě yǐ zǒu xià miàn nǎ tiáo lù xiàn
她可以走下面哪条路线？

- (A) → → ↑ ↑ → → ↑ ↑ → → →
- (C) → → ↑ ↑ ↑ ↑ → → → → →
- (E) → → ↑ ↑ ↑ → → ↑ → → →

- (B) → → ↑ ↑ → → → → → ↑ ↑
- (D) → → ↑ ↑ → → ↑ ↑ ← ← ←

第3课 逆向推导 (Work backwards)

《2013》 Pre-Ecolier 第2题

2. There are 12 books on a shelf and four children in a room. How many books will be left on the



shelf if each child takes one book?

(A) 12

(B) 8

(C) 4

(D) 2

(E) 0

《2015》 Level 1-2 第15题

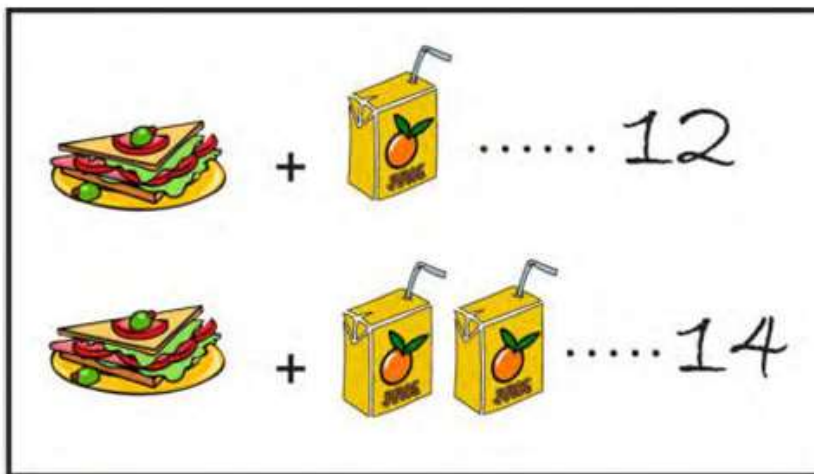
- 15.** Mother ordered 2 pizzas and sliced each of them into 8 pieces for Vera's birthday. There were 14 children at the party including Vera. How many slices are left over if mother gives one slice to each child? 妈妈为小V的生日宴会订了两张披萨并且将每张披萨都分成了8块。小V的生日宴会上有14个小小朋友（包括小V）。请问如果妈妈给每位小朋友一块披萨，那么还剩几块披萨？
- (A) 5 (B) 4 (C) 3 (D) 2 (E) 1

《2022》Level 1-2 第4题

4. One sandwich and one juice together cost 12 euro. One sandwich and two juices together cost 14 euro.

yī fèn sān míng zhì hé yī bēi guǒ zhī jiā qǐ lái gòng 12 yuán yī fèn sān míng zhì hé liǎng bēi guǒ
一份三明治和一杯果汁加起来共 12 元,一份三明治和两杯果

zhī jiā qǐ lái gòng 14 yuán
汁加起来共 14 元。



How much does one juice cost?

nà me yī bēi guǒ zhī duō shǎo qián
那么一杯果汁多少钱?

(A) 1

(B) 2

(C) 3

(D) 4

(E) 5

《2024》A级别 第24题

- 24、** Andrew was throwing darts at a target. He started with 10 darts and got 2 new darts each time he hit the target. In total, Andrew threw 20 darts and then had no darts left. How many times did Andrew hit the target?

*Andrew*正在朝着靶心投掷飞镖。一开始他有10个飞镖，每次击中靶心后他都会获得2个新飞镖。*Andrew*一共投掷了20个飞镖，并且最后没有剩下飞镖。请问*Andrew*一共击中了多少次靶心？

- A. 4 B. 5 C. 6 D. 8 E. 10

第4课 时间与日历 (Calendar and time)

《2012》 Pre-Ecolier 第6题

6. On Friday Dan starts to paint the word BANANA. Each day he paints one letter. On what day will he paint the last letter?

- (A) Monday (B) Tuesday (C) Wednesday (D) Thursday (E) Friday

《2012》 Pre-Ecolier 第11题

11. Today Betty added her age and her sister's age and obtained 10 as the sum. What will the sum of their ages be after one year?

(A) 5

(B) 10

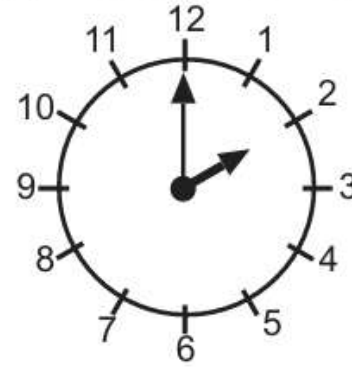
(C) 11

(D) 12

(E) 20

《2012》 Pre-Ecolier 第12题

12. The clock shows the time when Stephen leaves his school. School lunch starts 3 hours



before school ends. At what time does lunch start?

(A) 1

(B) 2

(C) 5

(D) 11

(E) 12

《2021》 Level 1-2 第22题

22. Kangie eats only apples on Monday, Wednesday and Friday. On Tuesdays and Thursdays he eats only mangoes. He eats either 2 apples or 3 mangoes a day. On Saturdays and Sundays he eats nothing. How many pieces of fruit does Kangie eat in two weeks?

zài zhōu yī zhōu sān hé zhōu wǔ zhǐ chī píng guǒ zài zhōu èr hé zhōu sì zhǐ chī
Kangie 在 周 一、周 三 和 周 五 只 吃 苹 果；在 周 二 和 周 四 只 吃
máng guǒ tā yì tiān chī gè píng guǒ huò gè máng guǒ qiě zài zhōu liù hé zhōu rì
芒 果。他 一 天 吃 2 个 苹 果 或 3 个 芒 果，且 在 周 六 和 周 日
shén me dōu bù chī qǐng wèn zài liǎng zhōu nèi chī le duō shǎo gè shuǐ guǒ
什 么 都 不 吃。请 问 Kangie 在 两 周 内 吃 了 多 少 个 水 果？

(A) 12

(B) 16

(C) 18

(D) 20

(E) 24

《2024》A级别 第3题

23. Julia has a list of problems to finish during May. She starts on the 1st of May, which this year is a Wednesday. If she solves exactly 2 problems each day, she will

finish the task on a Sunday. If she solves exactly 3 problems each day, she will finish the task on a Wednesday. How many problems are on the list?

*Julia*五月份需要完成一套试卷，他从五月一日开始做题。如果她每天恰好能解决2道题，她将在某个星期日完成整套试卷。如果她每天恰好能解决3道题，她将在某个星期三完成整套试卷。请问这套试卷一共有多少道题？



MAY 2024						
Mon	Tue	Wed	Thu	Fri	Sat	Sun
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

- A. 6 B. 12 C. 18 D. 24 E. 30

第5课 制作有序列表 (Organized list)

《2012》 Pre-Ecolier 第4题

4. Helena has written down the word KANGAROO twice. How many times did she write the letter A?

(A) 1

(B) 2

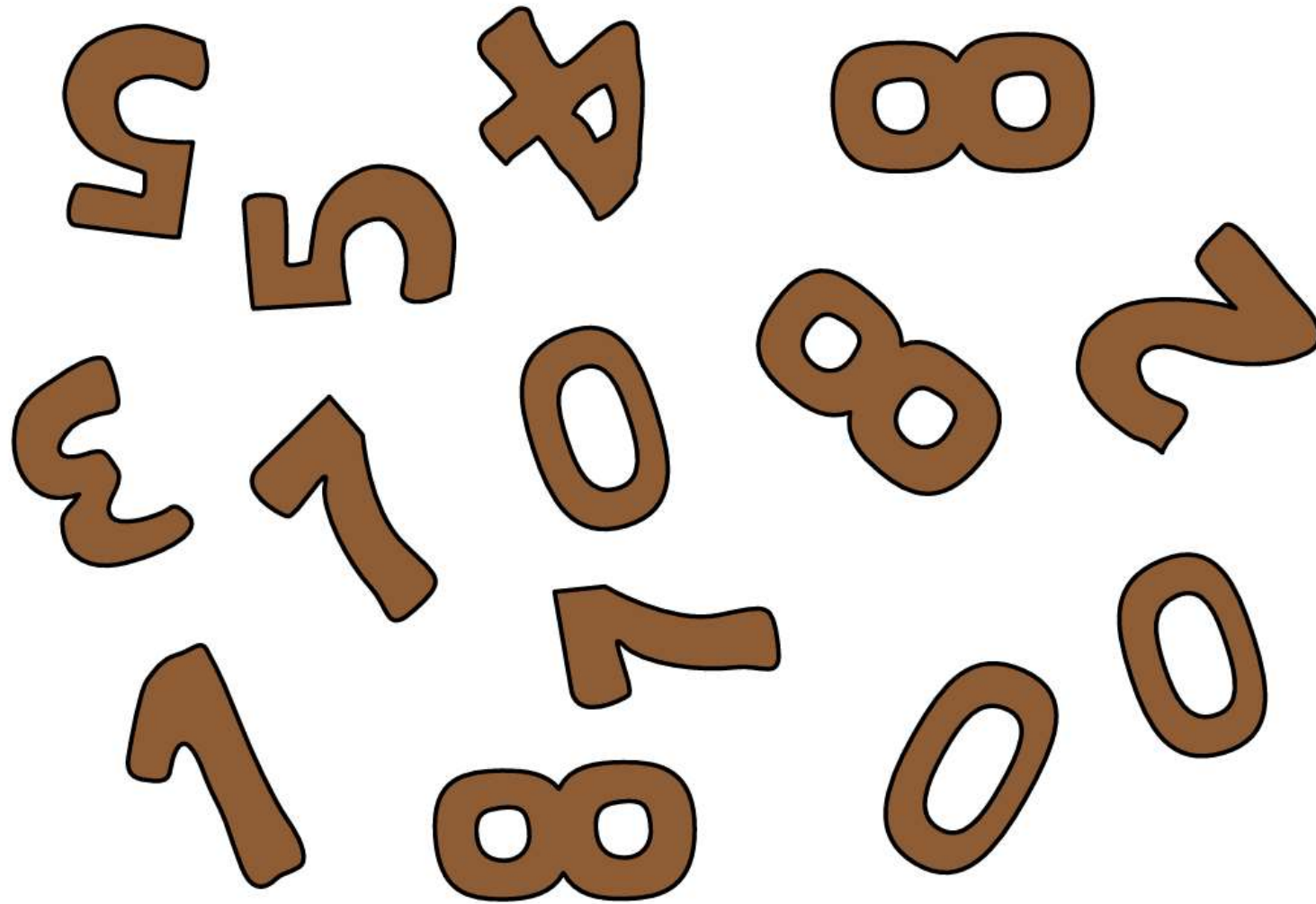
(C) 3

(D) 4

(E) 6

《2013》 Pre-Ecolier 第1题

1. Which digits are missing?



(A) 3 and 5

(B) 4 and 8

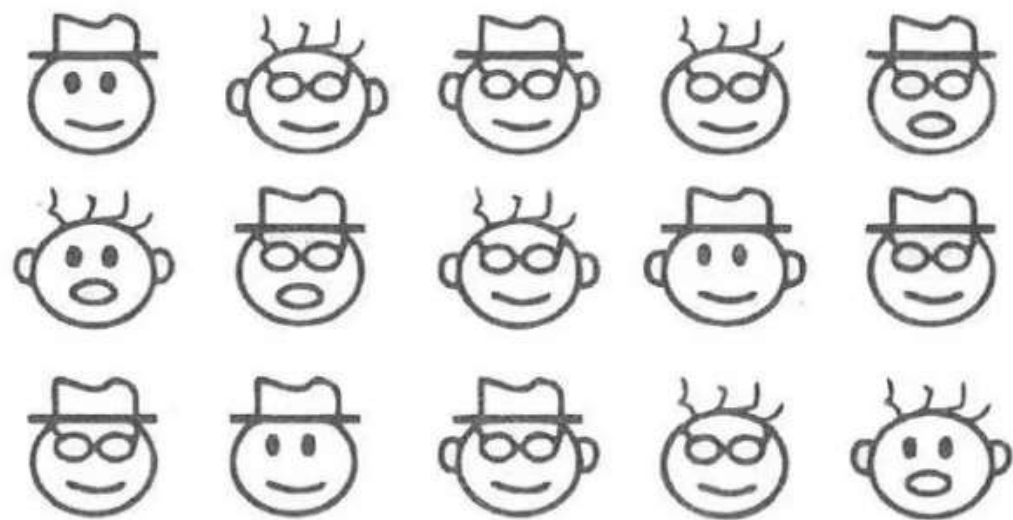
(C) 2 and 0

(D) 6 and 9

(E) 7 and 1

5. In the picture, there are 8 different faces.

如图所示，有8种不同的头像。



Each face appears twice, except for one. Which face appears only once?

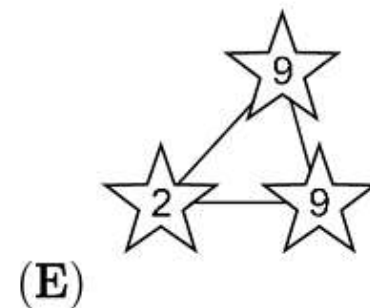
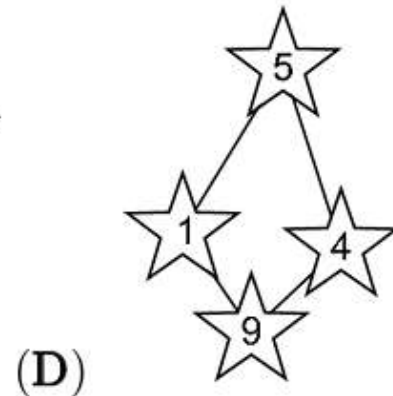
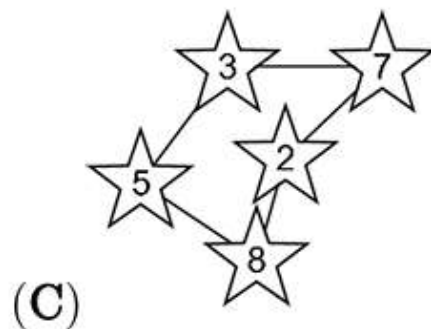
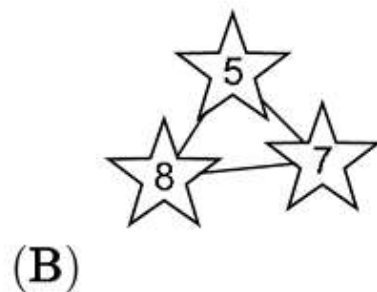
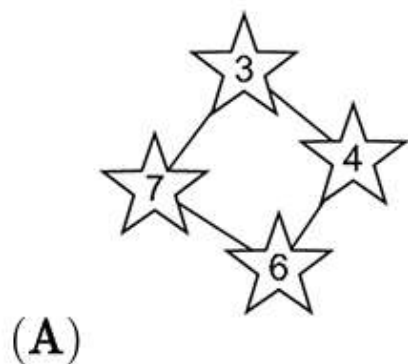
除了一个头像以外，其他每种头像出现两次。哪个头像只出现了一次？

- A.  B.  C.  D.  E. 

8. In the Kangaroo constellation, all stars have a number greater than 3 and their sum is 20.
Which is the Kangaroo constellation?

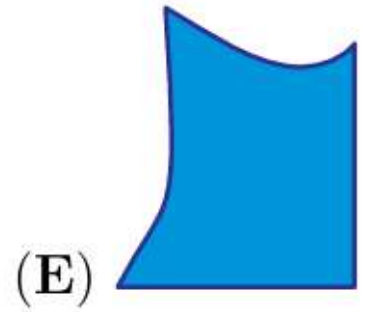
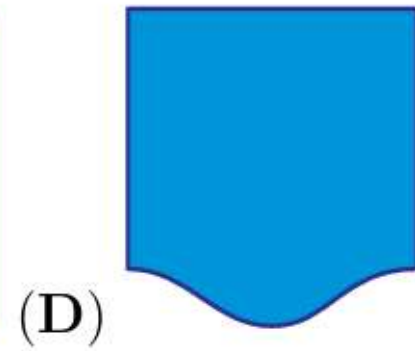
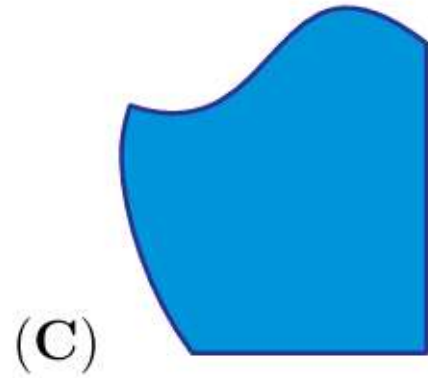
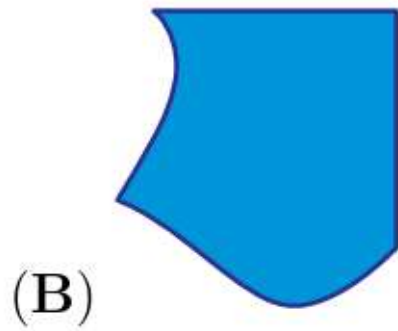
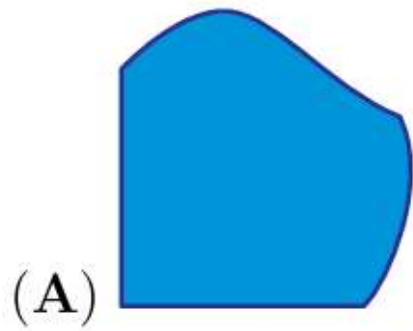
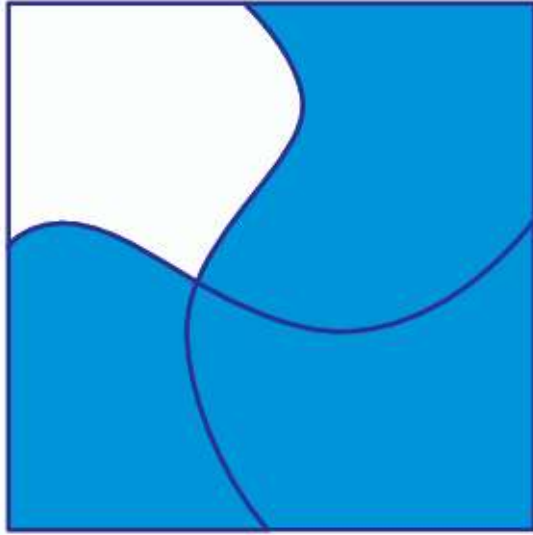
zài dài shǔ xīng zuò zhōng suǒ yǒu de xīng xīng shàng dōu yǒu yí gè dà yú de shù zì
在袋鼠星座中，所有的星星上都有一个大于3的数字，

ér qiě tā men de zǒng hé shì qǐng wèn nǎ gè shì dài shǔ xīng zuò
而且它们的总和是20。请问哪个是袋鼠星座？



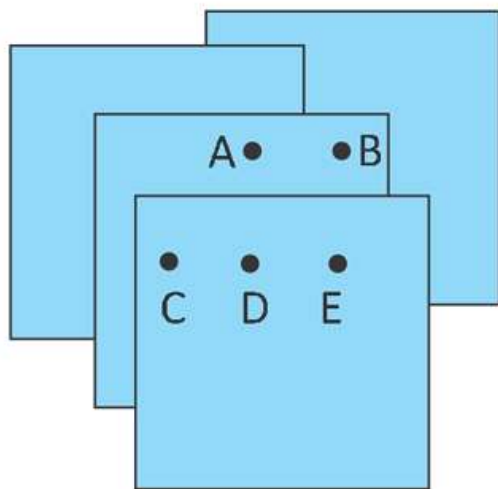
第6课 二维空间可视化 (2D visualization)

2. Which piece fits in the empty place?



2. Four identical pieces of paper are placed as shown. Michael wants to punch a hole that goes through all four pieces. At which point should Michael punch the hole?

jiāng sì zhāng xiāng tóng de zhǐ zhāng bǎi fàng rú xià xiǎng dǎ yí gè dòng jiāng sì
将四张相同的纸张摆放如下。Michael 想打一个洞将四
zhāng zhǐ quán bù chuān guò qǐng wèn yīng gāi bǎ dòng dǎ zài nǎ gè diǎn de wèi
张纸全部穿过。请问 Michael 应该把洞打在哪个点的位
zhì
置?



(A) A

(B) B

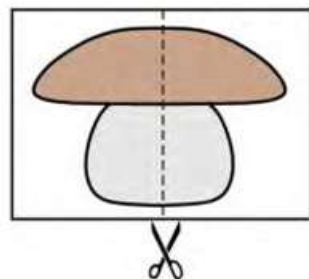
(C) C

(D) D

(E) E

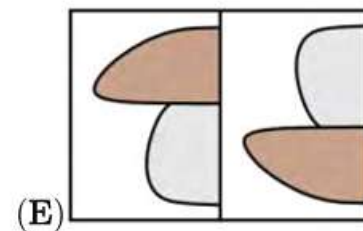
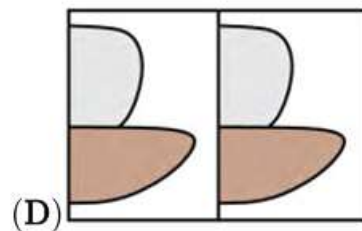
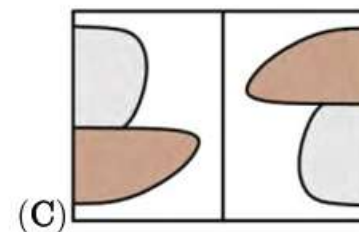
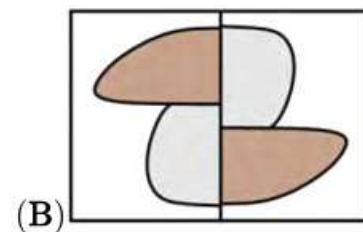
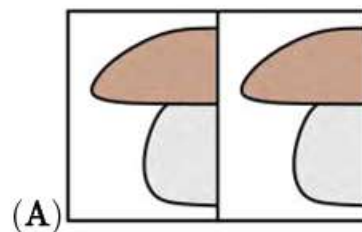
2. Arek cuts this picture in half and puts the two pieces together.

ā léi jiāng xià miàn zhè zhāng zhào piàn jiǎn chéng liǎng bàn , rán hòu chóng xīn pīn jiē zài yì qǐ
阿雷将下面这张照片剪成两半, 然后重新拼接在一起。



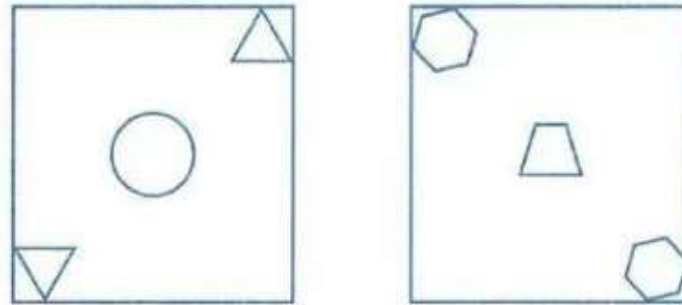
Which option shows the two pieces of Arek' s picture?

xià miàn nǎ ge xuǎn xiàng shì ā léi pīn jiē chū lái de xíng zhuàng ?
下面哪个选项是阿雷拼接出来的形状？



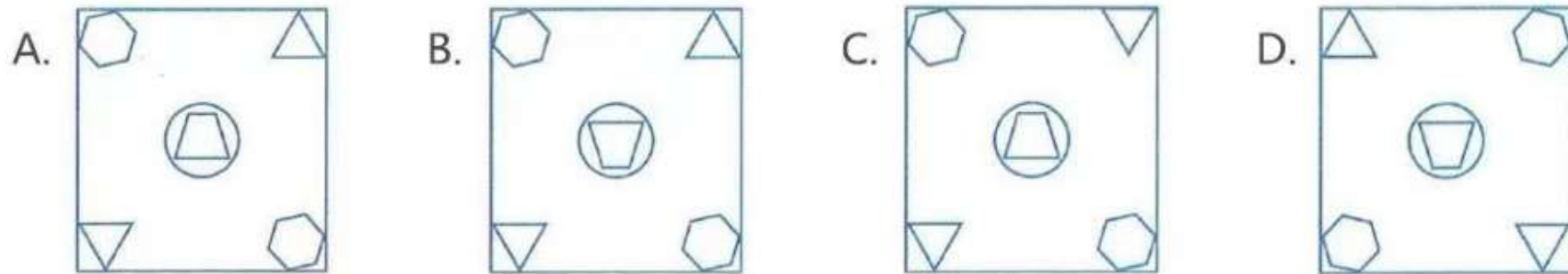
- 2、 Some shapes are printed on 2 pieces of glass.

两张透明卡片上印有一些图案.



Anna places one on top of the other, without turning either piece. What does she see?

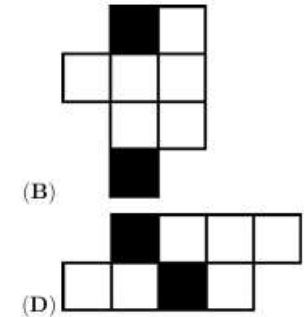
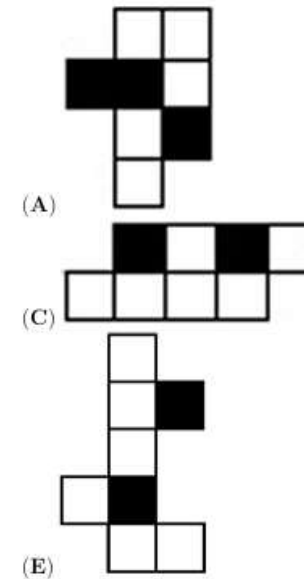
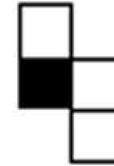
Anna将一个卡片放在另一个上面，如果不转动玻璃，她会看到什么图案？



15. Sepehr has two paperboards. He colored one side of each paperboard like the following. Which shape can he make using both paperboards?

Sepehr mempunyai dua keratan kertas. Dia mewarnakan satu permukaan pada setiap keratan tersebut seperti gambar di bawah. Bentuk yang manakah boleh dibuatnya dengan menggunakan kedua-dua keratan kertas tersebut?

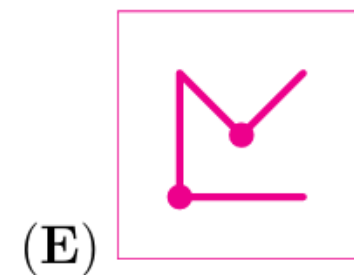
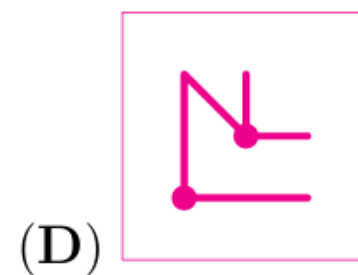
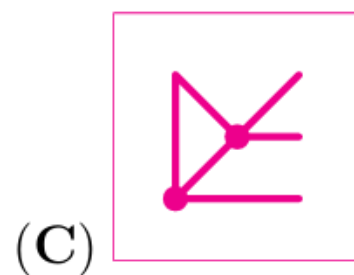
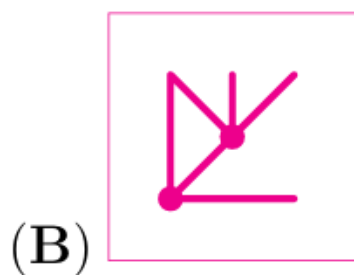
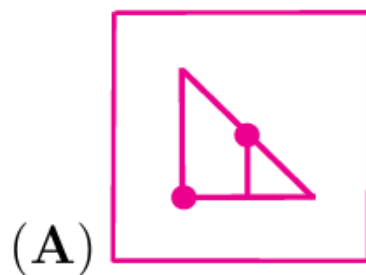
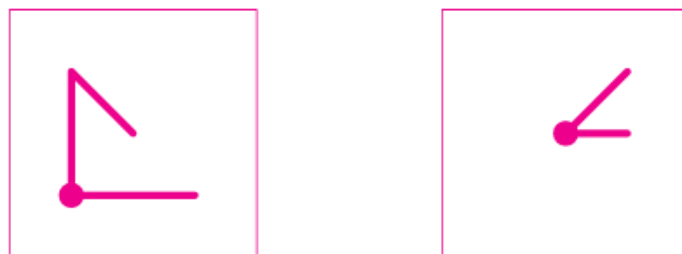
Sepehr 有两张卡片。他把每张卡片的其中一面涂成下图的样子。请问 he 可以用这两张卡片排成怎样的形状？



9. The two transparent squares are put on top of each other. What can you see?

Dua petak yang lutsinar ini disusun supaya salah satunya terletak di atas yang lain. Apakah yang dapat dilihat?

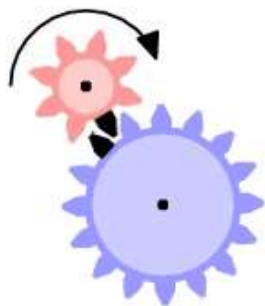
这两个透明的正方形被叠在一起。请问以下哪一个选择是你可以看到图形？



2021 Animation

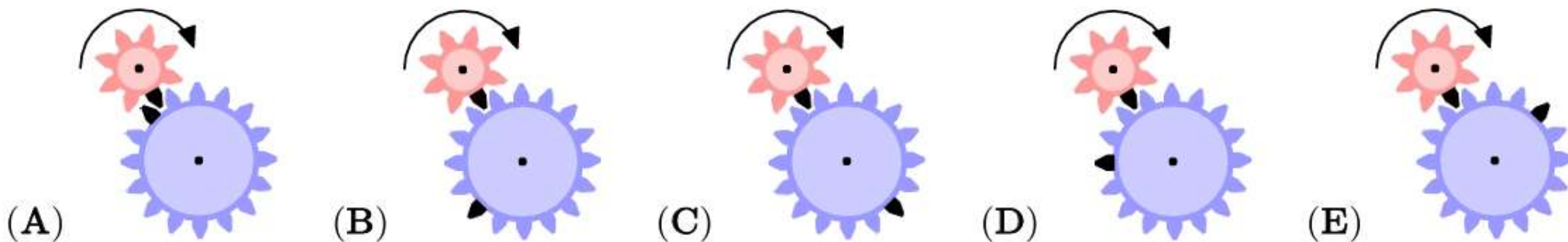
21. The picture shows two cogs, each with a black tooth.

xià tú zhōng yǒu liǎng gè chǐ lún měi gè chǐ lún shàng dōu yǒu yí gè hēi sè de lún chǐ
下图中有两个齿轮，每个齿轮上都有一个黑色的轮齿。



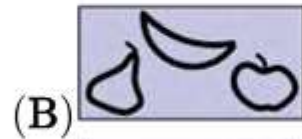
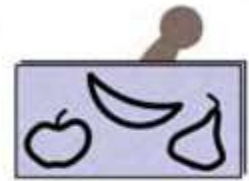
Where will the black teeth be after the small cog has made one full turn?

qǐng wèn xiǎo chǐ lún zhuàn dòng yì zhōu hòu hēi sè lún chǐ huì zài shén me wèi zhì
请问小齿轮转动一周后，黑色轮齿会在什么位置？



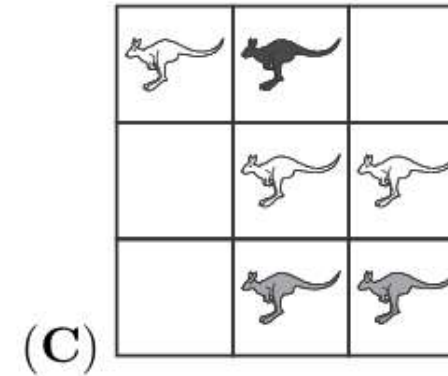
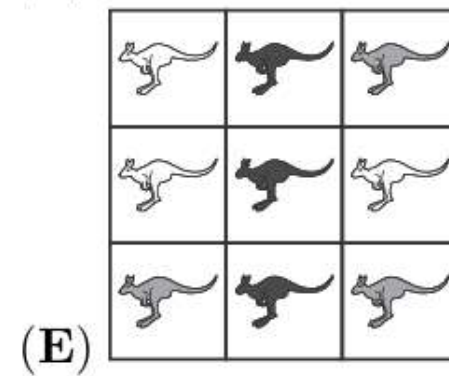
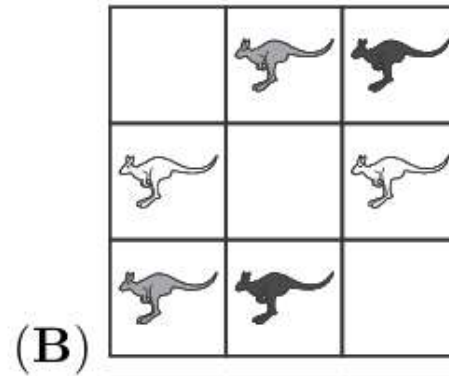
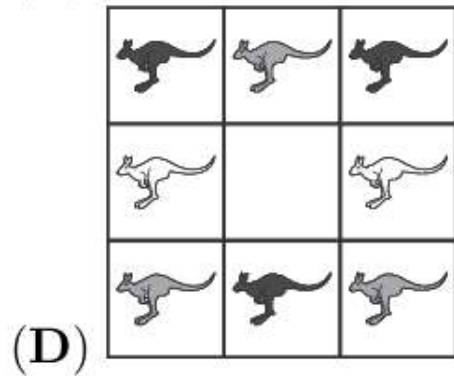
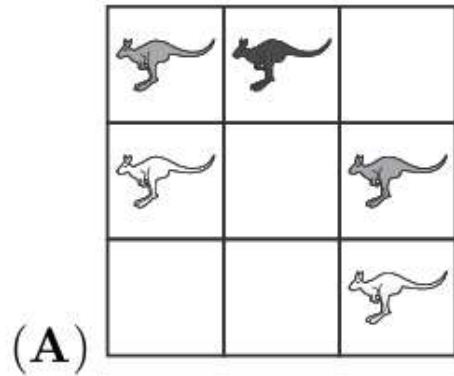
14. Which of the following pictures will we see when we use the stamp shown?

使用右图中的印章时,可以得到下面哪种印记?



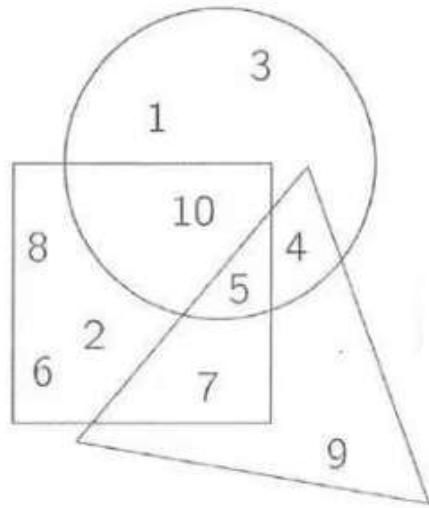
第7课 制作表格 (Make a table)

4. There are white, grey and black kangaroos. Which picture has more black kangaroos than white kangaroos?



1、 Which number is inside the triangle, the square, and the circle?

哪个数字同时在三角形、正方形和圆形里面？



A. 1

B. 4

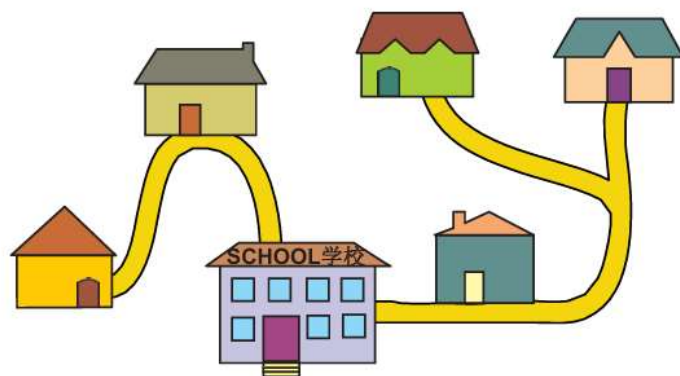
C. 5

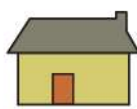
D. 9

E. 12

- # 15. The picture shows the five houses of five friends and their school. The school is the largest building in the picture. To go to school, Doris and Ali walk past Leo's house. Eva walks past Chole's house. Which is Eva's house?

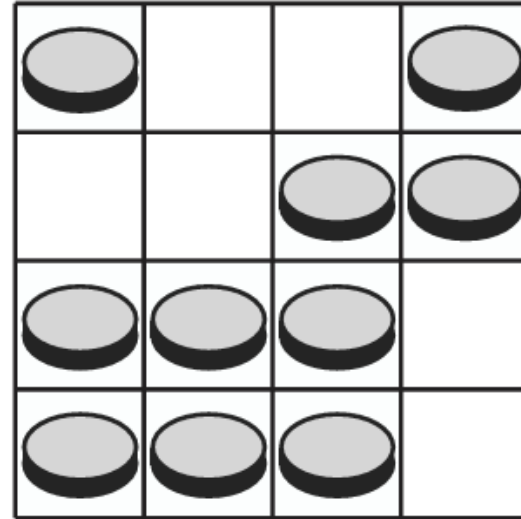
xià tú zhǎn shì le wǔ gè péng yǒu de jiā hé xué xiào de wèi zhì xué xiào shì tú zhōng zuì
 下图展示了五个朋友的家和学校的位置。学校是图中最
 dà de jiàn zhù shàng xué lù shàng hé huì jīng guò de jiā huì jīng guò
 大的建筑。上学路上 Doris 和 Ali 会经过 Leo 的家, Eva 会经过
 de jiā qǐng wèn nǎ yí gè shì de jiā
 Chole 的家。请问哪一个 是 Eva 的家?



- (A)  (B)  (C)  (D)  (E) 

第8课 猜想与验证 (Guess and check)

24. There are coins on the board. We want to have 2 coins in each column and 2 coins in



each row. How many coins need to be removed?

(A) 0

(B) 1

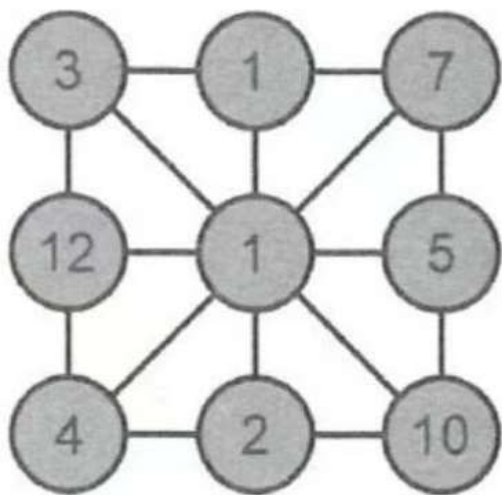
(C) 2

(D) 3

(E) 4

- 16、** One of the numbers in the picture is equal to the sum of the numbers connected directly to it. Which number is this?

下图中有一个数等于与它直接相连的所有数之和，请问是哪一个数？



- A. 3 B. 5 C. 7 D. 10 E. 12

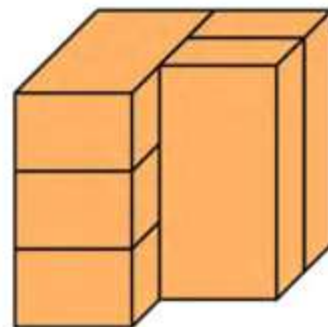
第9课 三维空间可视化 (3D visualization)

3. The picture shows 5 identical bricks.

How many bricks are touching exactly 3 other bricks?

tú zhōng yǒu kuài xiāng tóng de zhuān
图 中 有 5 块 相 同 的 砖 。

qí zhōng yǒu jǐ kuài zhuān qià hǎo āi zhe lìng wài 3 kuài zhuān
其 中 有 几 块 砖 恰 好 挨 着 另 外 3 块 砖 ？



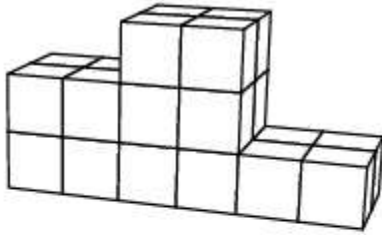
(A) 1

(B) 2

(C) 3

(D) 4

(E) 5



14.

Peter built a podium (as in the picture). How many cubes did he use?

(A) 12

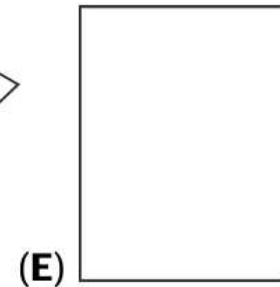
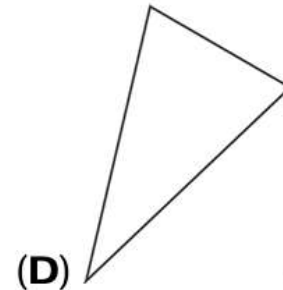
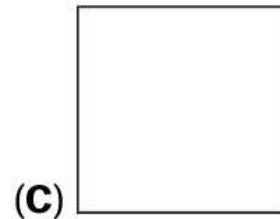
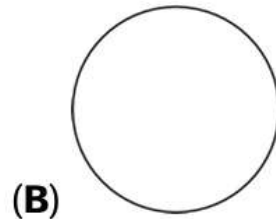
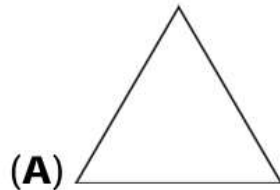
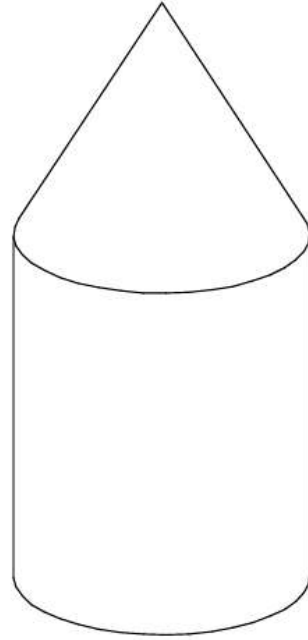
(B) 18

(C) 19

(D) 22

(E) 24

7. What does the round tower look like from above? 哪个图形是下面图形的俯视图?



13. Which option shows the view from above this stack of discs?

cóng shàng fāng kàn zhè dié yuán pán kě yǐ dé dào xià miàn nǎ fú tú
从上方看这叠圆盘可以得到下面哪幅图?

